Interfacing Data Transfer Models

Course Teacher:

Md. Obaidur Rahman, Ph.D.
Professor
Department of Computer Science and Engineering (CSE),
Dhaka University of Engineering & Technology (DUET), Gazipur.

Course ID: CSE - 4619
Course Title: Peripherals, Interfacing and Embedded Systems
Department of Computer Science and Engineering (CSE),
Islamic University of Technology (IUT), Gazipur.

Lecture References:

- **Book:**
  - *Microprocessor Architecture, Programming and Applications with 8085 (Part-III),* **Author:** Ramesh Gaonkor

- **Lecture Materials:**
  - *Ramesh Gaonkar*

Peripherals (I/Os) and Interfacing

- The primary functions of the microprocessor/microcontroller are:
  - to accept data from input devices such as – keyboards and A/D converters
  - read instructions from the memory
  - process data according to the instructions and
  - send the results to output devices such as – LEDs, printers and monitors.
- The input and output devices are called either **peripherals** or **I/O**.
- Designing logic circuits (hardware) and writing instructions (software) to enable microprocessor to communicate with I/Os is called **interfacing** and logic circuits are called **I/O ports** or **interfacing devices**.

Models of Data Transfer

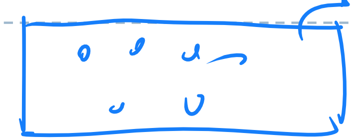
- Transmit or receive data occurs either in
- **Parallel Mode:**
  - The entire data (i.e., 8-bit or16-bit) is transferred at one time
  - In 8085, an 8-bit data is transferred simultaneously over the 8-data lines (i.e., data bus)
  - Connected via direct cable that has **one wire** for each bit in a character of data code being used by the terminal
  - With multiple wires, all the bits of a characters can be transmitted between the terminals and computer at once
  - **<u>Disadv:</u>** Very expensive & no practice over long distance
  - Devices use parallel mode: *Keyboards, 7-segment Display, Data Converters and Memory*.

Models of Data Transfer

- **Serial Mode:**
  - Bits of each character are sent down to a line **one by one**
  - Complicated process because machine needs to know how to **decompose** and to **reconstruct** of bits at each respective end
  - Data are transferred one bit at a time over a single line between the MP and a peripheral.
  - A data is converted to a stream of bits
  - Devices use serial mode: *CRT, Printers, USB, Modems, Telephone lines*.

Models of Data Transfer

port name DB9



RS232

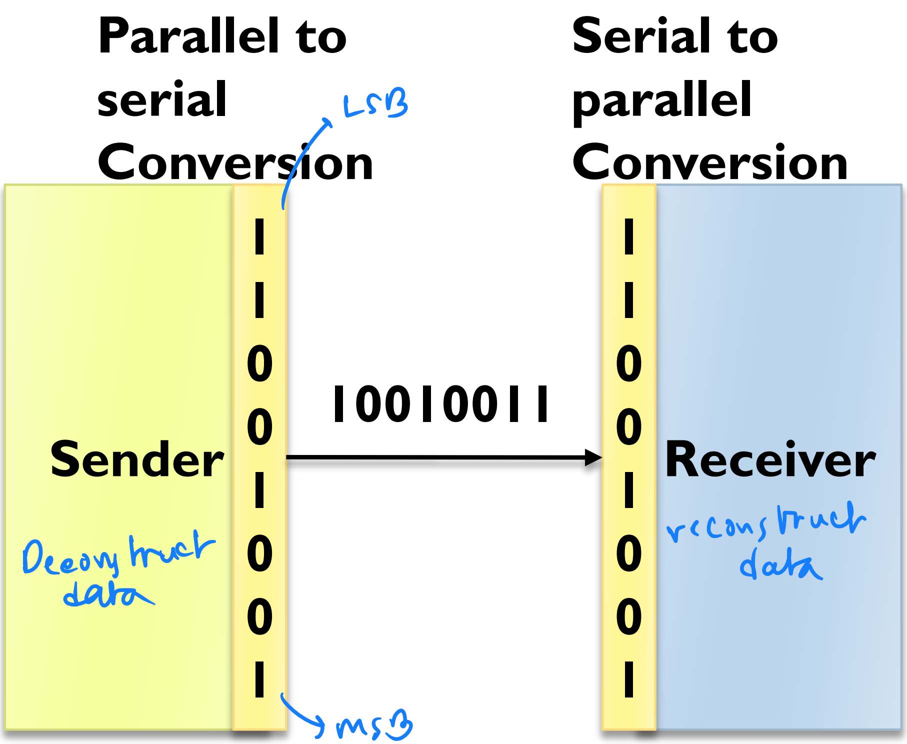
connects graphics card peripherals etc

- **Serial Mode:**
  - RS232- Typical computer COM port
  - SCI- Serial Communication interface, uses the universal asynchronous receiver/transmitter (USART) or UART
  - SPI Serial peripheral interface

USB has this

Models of Data Transfer

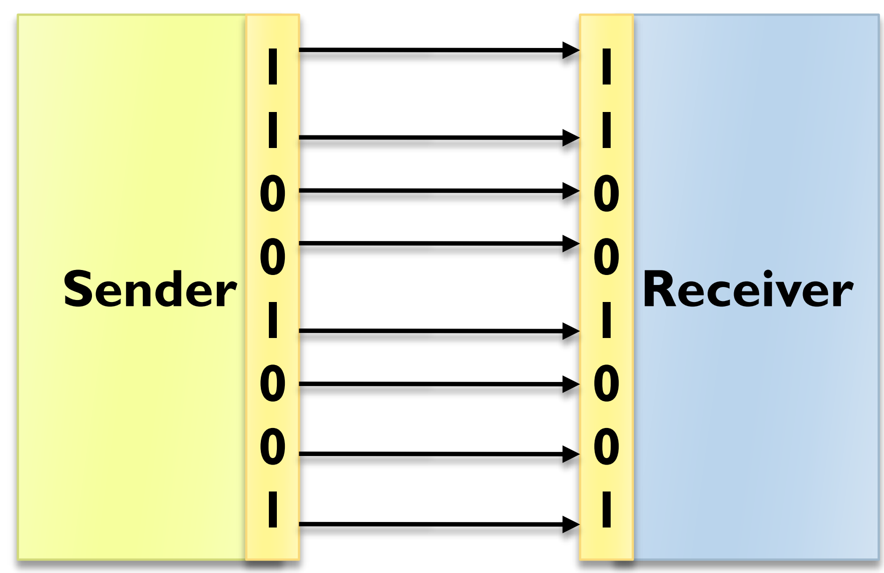
Two basic modes of data transmission



Serial Transmission

we send LSB first and build up to MSB

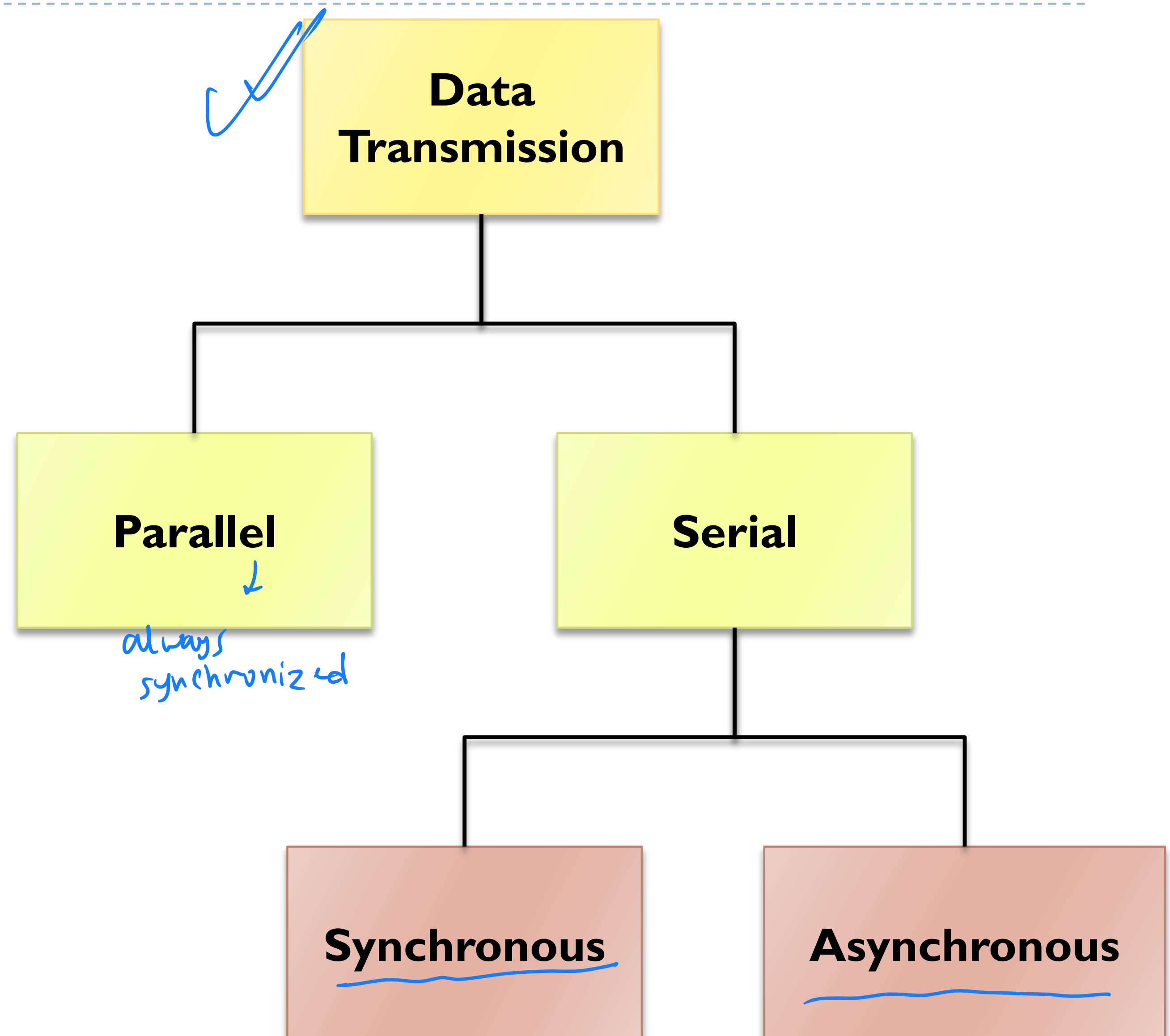
if one bit error occurs, we have to retransmit the entire data. Thus this is expensive



Parallel Transmission

Models of Data Transfer

- **Serial**
  - Cheaper
  - Slower
- **Parallel**
  - Faster
  - Limited to small distances
  - Simultaneous transmission
  - Requires separate data lines
  - Bits must stay synchronized
  - Expensive



Type of Serial Communication

- Microprocessor/microcontroller communicates with peripherals in either of two transmission formats:
- **Synchronous Transmission:**
  - Synchronous means at the same time
  - The transmit and receive are synchronized with same clock
  - Block of data can be sent
  - It is used for high-speed data transmission

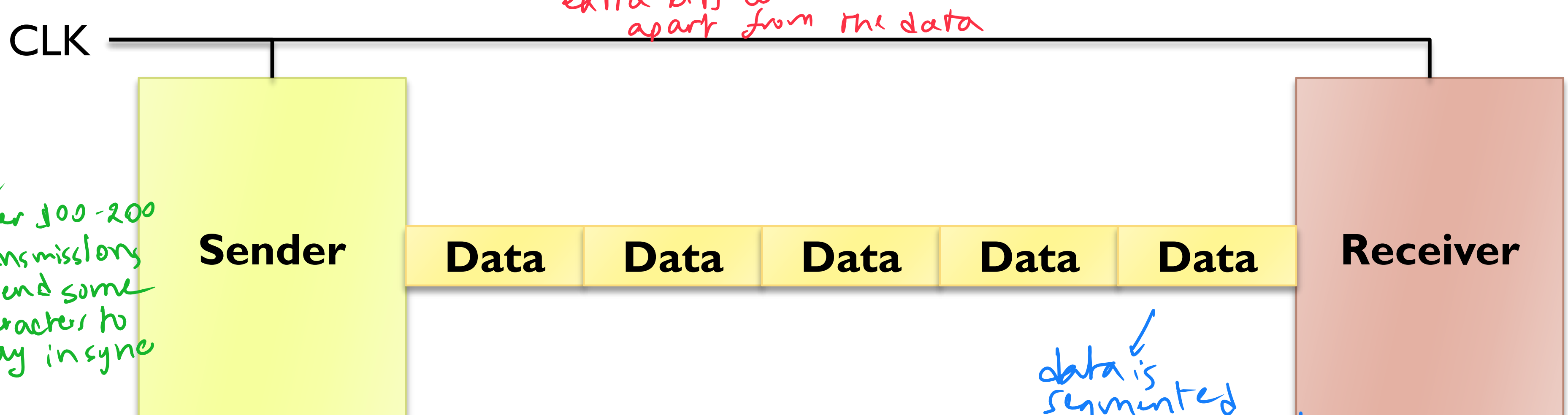
Type of Serial Communication

- **Synchronous Transmission:** (mostly used nowadays)
  - Data rates are dependent on clock rates
  - Continuously transmitting characters to remain in sync.
  - More efficient : Less overhead than asynchronous transmission

only overhead

to overcome clock skew problem (some clocks becoming slow and some becoming fast)

extra bits we have to send apart from the data



after 100-200 transmissions we send some characters to stay in sync

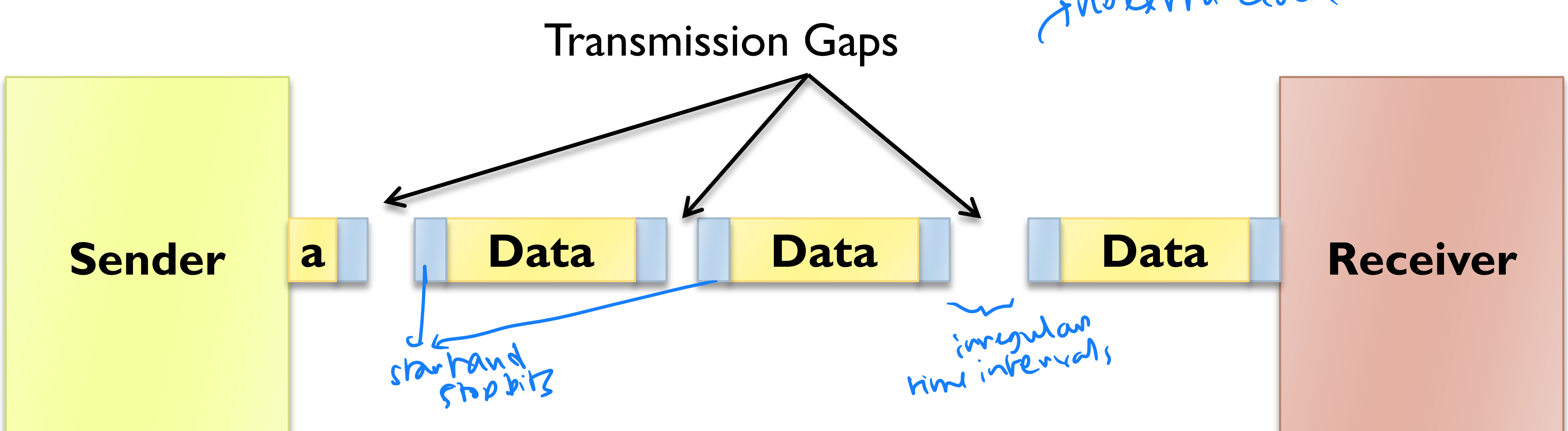
data is segmented into predefined data sizes

Synchronous transmission

Type of Serial Communication

- **Asynchronous Transmission:**
  - Asynchronous means irregular intervals
  - Each byte is encoded for transmission
    - Start and stop bits
  - No need for sender and receiver synchronization
  - It is used for low-speed data transmission
- Data transfer between microprocessor and peripherals are primarily ***asynchronous***. → in earlier days. now we use synchronous

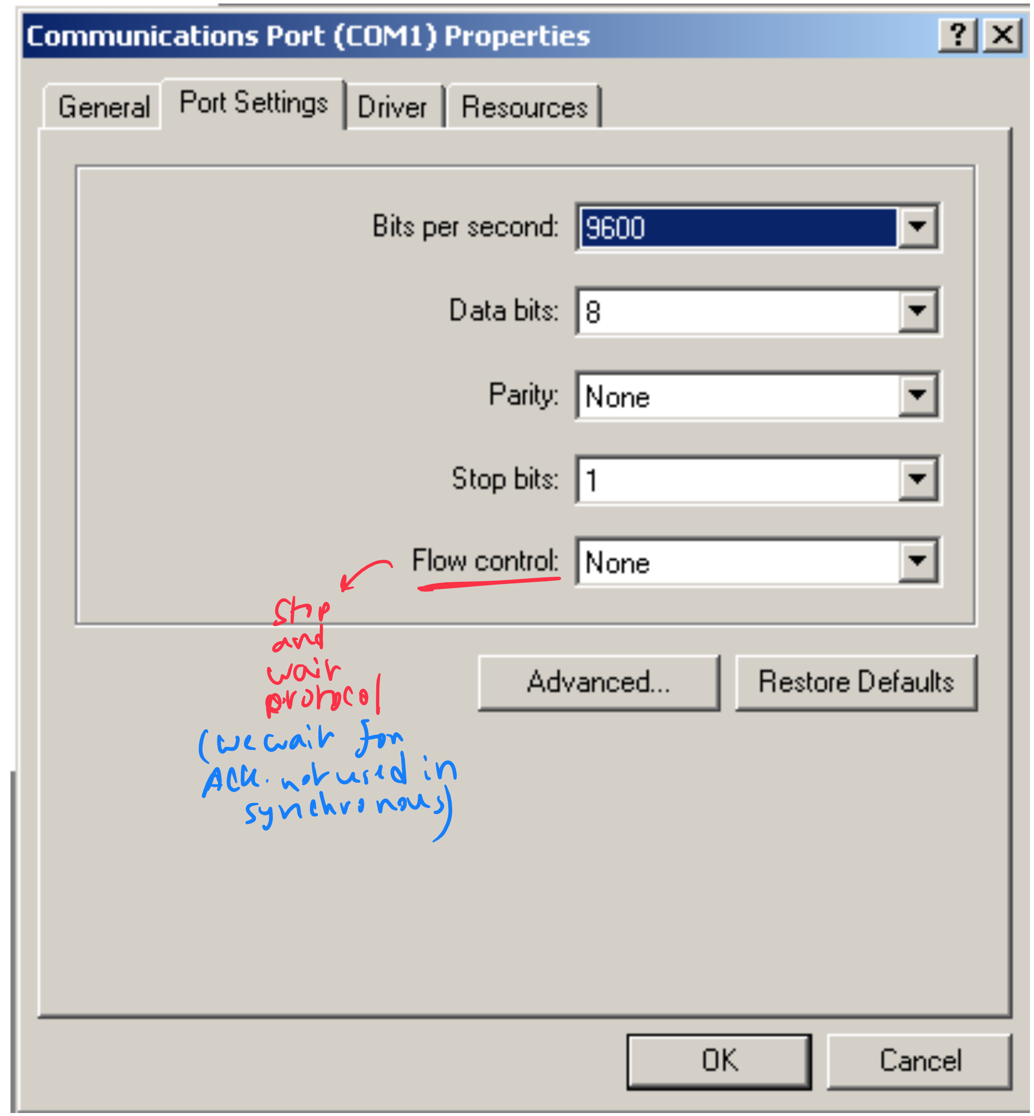
→ no extra clock



Asynchronous Serial Transmission

- Bits are transmitted in a specified format
- Defined by settings on transmitter and receiver:
  - Start Bit
  - Data Bits
  - Parity Bits
  - Stop Bits

for error checking



stop and wait protocol
(we wait for ACK. not used in synchronous)

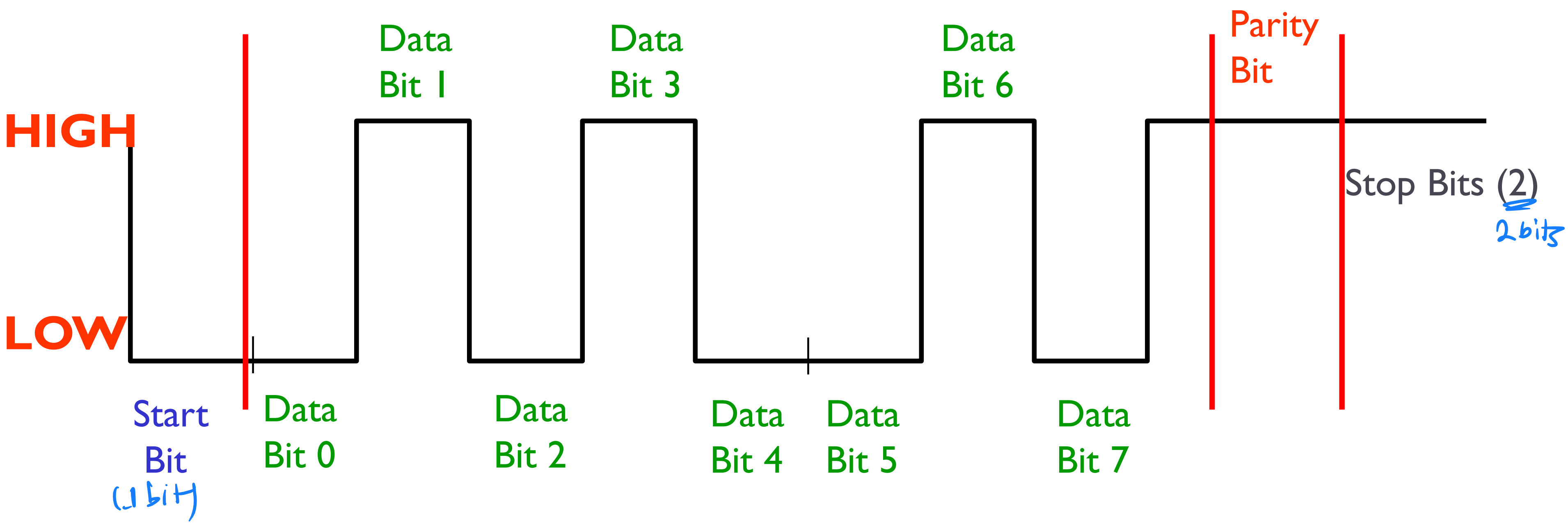
Example of Windows setting

Asynchronous Serial Transmission

One Data Package

- ♦ Four parts per package

start
stop
data
parity



Performance Comparison

- **Example Scenario**: Consider a character consists of
  01001011

- **Asynch**: 250 char x (8 data + 2 start/stop) = 2500 bits

- Assuming 1 ***Synch character*** after each 50 Character Transmission **Synch**: (250 + 5 synch char) x 8 bits) = 2040 bits

  each synch char has 8 bits → sent after every 50 transmission

- Thus, for the scenario Synch is approximately 20% more efficient than Aysnch

- **Note**: Mostly, host computers adopt Synch transmission.

New Trend: **Isochronous** Connections

- Apart from ***Synchronous*** and ***Asynchronous*** transmission a third type of connection defined at the data link layer used to support real-time applications
- Data must be delivered at just the right speed (real-time) – not too fast and not too slow
- Typically an ***Isochronous*** connection must allocate resources on both ends to maintain real-time communication
- USB and Firewire can both support isochronous connection
- Provide data transmission in a regular period of time

Conditions for Data Transfer

(almost everything is controlled by μp

- **Microprocessor Controlled Data Transfer**
  - ***Unconditional Data Transfer***
    - MP thinks of that a peripheral is always available
  - ***Data Transfer with Polling***
    - Polling a peripheral or set of peripherals regularly
  - ***Data Transfer with Interrupt***
    - A peripheral is ready to transfer data and sends an INT signal
    - MP stops current execution and send the peripheral an INTA signal
  - ***Data Transfer with READY Signal***
    - If peripherals response time is slower than the execution time of MP then T states are extended to complete data transfer → μp sends the ready signal and peripheral responds
  - ***Data Transfer with Handshake Signals***
    - Signal are exchanged between MP and peripheral prior to data transfer

Conditions for Data Transfer

- **Peripheral Controlled Data Transfer**
  - This type is applied when a peripheral is much faster than the MP
  - In case of Direct Memory Access (DMA), the DMA controller sends HOLD signal to the MP and MP releases its Data and Address Bus to the DMA controller by sending the HOLDA signal
  - Thereupon, data transferred at high-speed without the intervention of MP

Thank You !!